

Repository Intelligence Report

brody

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Statements are backed by repository evidence in the form path:line-range. Findings are labelled static analyzer (deterministic) or AI-inferred, and needs verification where evidence is incomplete.

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At a glance

brody: Repository intelligence, code review and code mapping.

brody is a full-stack monolith (Next.js) built with Next.js, React, organised into 26 functional areas. It exposes 33 API routes and stores 12 data models. It depends on Anthropic, OpenAI, SQLite, GitHub API.

174

Source files

48,377

Lines of code

TypeScript, CSS, Shell

Main languages

26

Functional areas

33

API routes

12

Data models

4

External services

0

Critical or high findings

Key points

What it is

brody: Repository intelligence, code review and code mapping.

How it is built

full-stack monolith (Next.js), using Next.js, React. 174 source files across TypeScript, CSS, Shell.

What it does

7 capabilities, led by API Layer, User Interface, Content Ingestion.

Where the risk is

No critical or high-severity issues were found; 25 lower-priority findings remain.

What to do next

Add tests for the code, which many files depend on.

What it does

API Layer: HTTP/RPC endpoints and request handling.

User Interface: Pages, components and client-side state.

Content Ingestion: Collecting and importing data from external sources.

Infrastructure & Deployment: Build, deployment and operations.

Application Core: Top-level application code.

AI Processing: Language-model prompts, agents and inference.

Deck: Code under `src/lib/deck/`.

Health and risk

No critical or high-severity issues were found; 25 lower-priority findings remain.

Strengths

- Automated tests exist (60 test files, Vitest, Playwright).
- A clear structure: full-stack monolith (Next.js).
- Configuration is read from environment variables.

Concerns

- 11 possible hard-coded credentials found in the code.

Recommended next steps

Now

Add tests for the code, which many files depend on.

This is the condensed report. The **Complete Technical Report** contains every finding, file, symbol and map in full.

1. Executive Summary

The executive summary is presented as slides at the front of this document.

Summary evidence

The in-depth summary that the executive summary above was distilled from. Each statement links to the code that supports it.

brody: Repository intelligence, code review and code mapping. (Stated in the project's own documentation.) (`package.json`)

The repository contains 174 source files (48,377 lines across 264 files) organised into 26 functional areas: API Layer, User Interface, Content Ingestion, Infrastructure & Deployment, Application Core, AI Processing.

Execution begins at `src/app/layout.tsx` (Next.js root layout (`package.json` script "dev")), `tools/build-worker.mjs` (`package.json` script "build"), `scripts/worker.mts` (`package.json` script "worker"). (`src/app/layout.tsx`, `tools/build-worker.mjs`, `scripts/worker.mts:8`)

Data enters through 33 API route(s) and 11 page(s) built with Next.js, React. Data is persisted in SQLite through Drizzle ORM, modelled by 12 entities such as projects, credentials, blobs, files. (`src/app/page.tsx:20`, `src/app/api/ai/models/route.ts:9`)

It integrates with Anthropic, OpenAI, SQLite, GitHub API. (`package.json`, `src/lib/config.ts:61`, `package.json`)

Engineering review found 25 finding(s): 0 critical, 0 high, 8 medium, 17 lower-priority. 25 came from deterministic analyzers and 0 from AI review. 60 test file(s) exist.

2. System at a Glance

Area	Description
Primary Purpose	Repository intelligence, code review and code mapping
Application Type	full-stack monolith (Next.js) (high confidence)
Primary Languages	TypeScript (223 files), CSS (1 files), Shell (2 files), SQL (2 files), JavaScript (3 files)
Frontend	Next.js, React, Tailwind CSS
Backend	custom HTTP handlers
Data Layer	Database: SQLite; Access: Drizzle ORM; 12 modelled entities
Authentication	no authentication mechanism detected
External Services	Anthropic, OpenAI, SQLite, GitHub API
Infrastructure	GitHub Actions, Docker, Docker Compose
Testing	60 test files; Vitest, Playwright

3. Architecture Overview

The codebase is best described as a full-stack monolith (Next.js) (high confidence). Both frontend and backend frameworks are present in one codebase.

Files divide into these layers: Entry Points (6), Presentation / UI (18), API / Transport (40), Domain / Services (96), Background Processing (2), AI (11), Data (4), Infrastructure & Config (11), Tests & Docs (66).

The largest functional areas are Testing (60 files), API Layer (27 files), Documentation (17 files), User Interface (15 files), Content Ingestion (13 files).

Layers: Entry Points (6 files) · Presentation / UI (18 files) · API / Transport (40 files) · Domain / Services (96 files) ·

4. Primary Runtime Flow

GET /api/projects/:id/files/content is handled by GET /api/projects/:id/files/content, which calls guard, getReadyProject, AppError, getDb, getFileContent; along the way it reads files, symbols, relationships, findings. (src/app/api/projects/[id]/files/content/route.ts:15)

1. **GET /api/projects/:id/files/content** — function at src/app/api/projects/[id]/files/content/route.ts:15 (src/app/api/projects/[id]/files/content/route.ts:15)
2. **guard** — function at src/lib/api.ts:20 (src/lib/api.ts:20)
3. **getReadyProject** — function at src/lib/api.ts:34 (src/lib/api.ts:34)
4. **AppError** — class at src/lib/util/errors.ts:2 (src/lib/util/errors.ts:2)
5. **getDb** — function at src/lib/db/client.ts:55 (src/lib/db/client.ts:55)
6. **getFileContent** — function at src/lib/ingest/store.ts:86 (src/lib/ingest/store.ts:86)
7. **changelImpact** — function at src/lib/map/index.ts:371 (src/lib/map/index.ts:371)
8. **json** — function at src/lib/api.ts:8 (src/lib/api.ts:8)
9. **chunks** — function at src/app/api/projects/[id]/files/content/route.ts:13 (src/app/api/projects/[id]/files/content/route.ts:13)
10. **reads files** — model at src/lib/db/schema.ts:42 (src/lib/db/schema.ts:42)
11. **reads symbols** — model at src/lib/db/schema.ts:82 (src/lib/db/schema.ts:82)
12. **reads relationships** — model at src/lib/db/schema.ts:110 (src/lib/db/schema.ts:110)
13. **reads findings** — model at src/lib/db/schema.ts:128 (src/lib/db/schema.ts:128)
14. **fail** — function at src/lib/api.ts:12 (src/lib/api.ts:12)

5. Major Functional Areas

Each area is a capability delivered by several files working together. This is the macro view; folders are covered next and single files further below.

Area	What it does	Files	Used by
Testing	Automated tests, fixtures and mocks.	60	Infrastructure & Deployment, Parse, API Layer, Analysis, Data Layer and 9 more
API Layer	HTTP/RPC endpoints and request handling.	27	Testing
Documentation	Project documentation.	17	API Layer, Testing, Export, Deck, Formal and 18 more
User Interface	Pages, components and client-side state.	15	API Layer, File & Media Storage, Reporting & Analytics, Map
Content Ingestion	Collecting and importing data from external sources.	13	Testing, API Layer, Infrastructure & Deployment, Background Jobs, File & Media Storage and 8 more
Infrastructure & Deployment	Build, deployment and operations.	12	Testing, Parse, Deck, Export, Discover and 17 more
Application	Top-level application code.	11	No other area depends on it directly
Core AI Processing	Language-model prompts, agents and inference.	11	Testing, API Layer, Configuration, Review, Documentation and 20 more
Deck	Code under src/lib/deck/.	11	Testing, API Layer, Export, Background Jobs, Graph and 1 more
Parse	Code under src/lib/parse/.	10	Testing, Graph, Analysis, Formal, Review and 1 more
Shared Utilities	Reusable helpers and shared code.	8	API Layer, Testing, User Interface, File & Media Storage, Content Ingestion and 14 more

Area	What it does	Files	Used by
Export	Code under <code>src/lib/export/</code> .	7	Testing, Infrastructure & Deployment, API Layer, Bundle, Deck and 3 more
Util	Code under <code>src/lib/util/</code> .	7	API Layer, Content Ingestion, Testing, Documentation, Review and 14 more
Configuration	Runtime configuration and tooling settings.	6	Documentation, Testing
Public	Code under <code>public/</code> .	5	No other area depends on it directly
Background Jobs	Asynchronous and scheduled work.	5	Testing, Infrastructure & Deployment, API Layer, File & Media Storage, Application Core and 1 more
Review	Code under <code>src/lib/review/</code> .	5	Testing, Analysis, Formal, Background Jobs, Deck and 1 more
Data Layer	Schemas, models, migrations and data access.	4	Testing, API Layer, Search & Retrieval, Documentation, Background Jobs and 17 more
File & Media Storage	Uploads and stored media.	4	Testing
Analysis	Code under <code>src/lib/analysis/</code> .	4	Testing, Review, Parse, Graph, Deck and 1 more
Discover	Code under <code>src/lib/discover/</code> .	4	Testing, API Layer, Documentation, Map, Review and 6 more
Formal	Code under <code>src/lib/formal/</code> .	4	Testing, API Layer, Review
Map	Code under <code>src/lib/map/</code> .	3	API Layer, Testing, Export, User Interface, Background Jobs and 4 more
Drizzle	Code under <code>drizzle/</code> .	2	No other area depends on it directly
Reporting & Analytics	Aggregations, reporting and analytics.	2	Testing
Search & Retrieval	Indexing and querying content.	2	Testing, Ask, Documentation, Bundle, Infrastructure & Deployment and 3 more
Bundle	Code under <code>src/lib/bundle/</code> .	2	Testing, API Layer, File & Media Storage
Graph	Code under <code>src/lib/graph/</code> .	2	Testing, Documentation, Review, Discover, Analysis and 2 more
Ask	Code under <code>src/lib/ask/</code> .	1	API Layer, Infrastructure & Deployment, Testing

Folders:

Folder	Files	What it is for
<code>drizzle/</code>	3	2 configuration, 1 schema. It spans the Data Layer, Drizzle functional areas.
<code>drizzle/meta/</code>	2	2 configuration. It sits inside the Drizzle functional area.
<code>fixtures/sample-shop/</code>	4	4 test. It sits inside the Testing functional area.
<code>scripts/</code>	7	4 service-layer, 2 utility, 1 entry-point. It sits inside the Infrastructure & Deployment functional area.
<code>src/</code>	162	67 service-layer, 40 API-layer, 18 user-interface. It spans the Data Layer, Configuration, API Layer and 22 more functional...
<code>src/app/</code>	41	37 API-layer, 1 schema, 1 utility. It spans the Data Layer, Configuration, API Layer and 6 more functional areas. It serves GET...
<code>src/app/api/</code>	27	26 API-layer, 1 schema. It spans the Data Layer, Configuration, API Layer and 4 more functional areas. It serves GET...
<code>src/app/api/ai/</code>	2	1 schema, 1 API-layer. It spans the Data Layer, Configuration functional areas. It serves GET <code>/api/ai/models</code> , GET...
<code>src/app/api/jobs/[id]/</code>	2	2 API-layer. It sits inside the Background Jobs functional area. It serves GET <code>/api/jobs/:id</code> , POST <code>/api/jobs/:id/cancel</code> .

Folder	Files	What it is for
src/app/api/projects/	21	21 API-layer. It spans the API Layer, File & Media Storage, Reporting & Analytics and 1 more functional areas. It serves GET...
src/app/api/projects/[id]/	17	17 API-layer. It spans the API Layer, File & Media Storage, Reporting & Analytics and 1 more functional areas. It serves DELETE...
src/app/api/projects/[id]/files/	2	2 API-layer. It sits inside the File & Media Storage functional area. It serves GET /api/projects/:id/files, GET...
src/app/p/[id]/	11	10 API-layer, 1 user-interface. It spans the API Layer, File & Media Storage, User Interface and 1 more functional areas. It...
src/components/	14	14 user-interface. It sits inside the User Interface functional area.

6. Data Flow

GET /path

```

GET /path
  ↓
lineAt
  ↓
callArgs
  ↓
isInlineFn
  ↓
reads symbols
  ↓
jobs.push
  ↓
split
  ↓
trim

```

GET /path is handled by GET /path, which calls lineAt, callArgs, isInlineFn, jobs.push, split; along the way it reads symbols. (src/lib/discover/detect.ts:315)

Data touched: symbols; external: none.

GET /api/ai/models

```

GET /api/ai/models
  ↓
guard
  ↓
AppError
  ↓
json
  ↓
listModels
  ↓
fail
  ↓
resolveModel
  ↓
providerConfigured
  ↓
explainAIError
  ↓
cacheKey
  ↓
split
  ↓
fetchCatalogue

```

```
↓  
isAppError  
↓  
scrubToken
```

GET /api/ai/models is handled by GET /api/ai/models, which calls guard, AppError, json, listModels, fail. (src/app/api/ai/models/route.ts:9)

GET /api/ai/settings

```
GET /api/ai/settings  
↓  
guard  
↓  
json  
↓  
view  
↓  
fail  
↓  
readSettings  
↓  
providerStatus  
↓  
providerConfigured  
↓  
resolveModel  
↓  
runtimeKind  
↓  
resolveEmbeddingModel  
↓  
endpoint  
↓  
selectedProvider  
↓  
isAppError
```

GET /api/ai/settings is handled by GET /api/ai/settings, which calls guard, json, view, fail, readSettings. (src/app/api/ai/settings/route.ts:38)

PUT /api/ai/settings

```
PUT /api/ai/settings  
↓  
guard  
↓  
AppError  
↓  
json  
↓  
updateSettings  
↓  
resetProviderCache  
↓  
checkProvider  
↓  
view  
↓  
fail  
↓  
readSettings  
↓  
settingsPath  
↓  
withDeadline
```

```
↓
explainAIError
↓
providerStatus
```

PUT /api/ai/settings is handled by PUT /api/ai/settings, which calls guard, AppError, json, updateSettings, resetProviderCache. (src/app/api/ai/settings/route.ts:43)

POST /api/github/validate

```
POST /api/github/validate
↓
guard
↓
json
↓
AppError
↓
parseGitHubUrl
↓
fetchGitHubMetadata
↓
trim
↓
fail
↓
split
↓
isAppError
↓
scrubToken
```

POST /api/github/validate is handled by POST /api/github/validate, which calls guard, json, AppError, parseGitHubUrl, fetchGitHubMetadata. (src/app/api/github/validate/route.ts:10)

GET /api/jobs/:id

```
GET /api/jobs/:id
↓
guard
↓
getJob
↓
AppError
↓
json
↓
fail
↓
getDb
↓
reads jobs
↓
isAppError
↓
scrubToken
↓
openDatabase
```

GET /api/jobs/:id is handled by GET /api/jobs/:id, which calls guard, getJob, AppError, json, fail; along the way it reads jobs. (src/app/api/jobs/[id]/route.ts:8)

Data touched: jobs; external: none.

POST /api/jobs/:id/cancel

```
POST /api/jobs/:id/cancel
↓
guard
↓
requestCancel
↓
AppError
↓
json
↓
fail
↓
getDb
↓
getJob
↓
isAppError
↓
scrubToken
↓
openDatabase
↓
reads jobs
```

POST /api/jobs/:id/cancel is handled by POST /api/jobs/:id/cancel, which calls guard, requestCancel, AppError, json, fail; along the way it reads jobs. (src/app/api/jobs/[id]/cancel/route.ts:8)

Data touched: jobs; external: none.

GET /api/projects

```
GET /api/projects
↓
guard
↓
getDb
↓
json
↓
reads projects
↓
fail
↓
openDatabase
↓
isAppError
↓
scrubToken
↓
migrationsFolder
```

GET /api/projects is handled by GET /api/projects, which calls guard, getDb, json, fail, openDatabase; along the way it reads projects. (src/app/api/projects/route.ts:7)

Data touched: projects; external: none.

DELETE /api/projects/:id

```
DELETE /api/projects/:id
↓
guard
↓
getProject
↓
```

```
listJobs
  ↓
requestCancel
  ↓
deleteProject
  ↓
json
  ↓
fail
  ↓
getDb
  ↓
AppError
  ↓
reads projects
  ↓
reads jobs
  ↓
getJob
  ↓
isAppError
```

DELETE /api/projects/:id is handled by DELETE /api/projects/:id, which calls guard, getProject, listJobs, requestCancel, deleteProject; along the way it reads projects, jobs. (src/app/api/projects/[id]/route.ts:15)

Data touched: projects, jobs; external: none.

GET /api/projects/:id

```
GET /api/projects/:id
  ↓
guard
  ↓
json
  ↓
projectSummary
  ↓
getProject
  ↓
fail
  ↓
getDb
  ↓
reads jobs
  ↓
AppError
  ↓
reads projects
  ↓
isAppError
  ↓
scrubToken
  ↓
openDatabase
```

GET /api/projects/:id is handled by GET /api/projects/:id, which calls guard, json, projectSummary, getProject, fail; along the way it reads jobs, projects. (src/app/api/projects/[id]/route.ts:8)

Data touched: jobs, projects; external: none.

POST /api/projects/:id/analyze

```
POST /api/projects/:id/analyze
  ↓
guard
  ↓
getProject
```

```
↓
getDb
↓
decryptSecret
↓
parseGitHubUrl
↓
fetchGitHubMetadata
↓
createPendingProject
↓
enqueueAnalysis
↓
json
↓
projectSummary
↓
reads credentials
↓
reads projects
↓
fail
```

POST /api/projects/:id/analyze is handled by POST /api/projects/:id/analyze, which calls guard, getProject, getDb, decryptSecret, parseGitHubUrl; along the way it reads credentials, projects. (src/app/api/projects/[id]/analyze/route.ts:17)

Data touched: credentials, projects; external: none.

GET /api/projects/:id/architecture

```
GET /api/projects/:id/architecture
↓
guard
↓
getReadyProject
↓
json
↓
architectureDiagramText
↓
architectureMermaid
↓
erMermaid
↓
legendText
↓
fail
↓
AppError
↓
getProject
↓
glyphFor
↓
loadModel
↓
box
```

GET /api/projects/:id/architecture is handled by GET /api/projects/:id/architecture, which calls guard, getReadyProject, json, architectureDiagramText, architectureMermaid. (src/app/api/projects/[id]/architecture/route.ts:8)

GET /api/projects/:id/ask

```
GET /api/projects/:id/ask
↓
guard
↓
getReadyProject
↓
json
↓
recentQuestions
↓
fail
↓
AppError
↓
getProject
↓
getDb
↓
reads questions
↓
isAppError
↓
scrubToken
↓
reads projects
↓
openDatabase
```

GET /api/projects/:id/ask is handled by GET /api/projects/:id/ask, which calls guard, getReadyProject, json, recentQuestions, fail; along the way it reads questions, projects. (src/app/api/projects/[id]/ask/route.ts:11)

Data touched: questions, projects; external: none.

POST /api/projects/:id/ask

```
POST /api/projects/:id/ask
↓
guard
↓
getReadyProject
↓
json
↓
AppError
↓
askRepository
↓
fail
↓
getProject
↓
getDb
↓
getAIProvider
↓
retrieveContext
↓
queryTerms
↓
UsageMeter
↓
tryAnalyze
```

POST /api/projects/:id/ask is handled by POST /api/projects/:id/ask, which calls guard, getReadyProject, json,

AppError, askRepository. (src/app/api/projects/[id]/ask/route.ts:19)

GET /api/projects/:id/deck

```
GET /api/projects/:id/deck
  ↓
guard
  ↓
getReadyProject
  ↓
json
  ↓
deckOutline
  ↓
isDeckFormat
  ↓
AppError
  ↓
exportDeck
  ↓
htmlSecurityHeaders
  ↓
fail
  ↓
getProject
  ↓
ensureDeckContent
  ↓
layoutDeck
  ↓
getDb
```

GET /api/projects/:id/deck is handled by GET /api/projects/:id/deck, which calls guard, getReadyProject, json, deckOutline, isDeckFormat. (src/app/api/projects/[id]/deck/route.ts:14)

7. API Architecture

44 route(s) were detected across Next.js, Express-style: 33 API and 11 page routes. (src/app/page.tsx:20, src/app/api/ai/models/route.ts:9, src/app/api/ai/settings/route.ts:38)

0 route(s) show an authentication or authorization check near their definition; 33 API route(s) show none, which means either the check lives in shared middleware or the route is open.

Method	Route	Handler	Authentication	Kind	Location
GET	/	uploadWithProgress	Not visible	page · Next.js	src/app/page.tsx:20
GET	/api/ai/models	GET (route handler)	Not visible	api · Next.js	src/app/api/ai/models/route.ts:9
GET	/api/ai/settings	GET (route handler)	Not visible	api · Next.js	src/app/api/ai/settings/route.ts:38
PUT	/api/ai/settings	PUT (route handler)	Not visible	api · Next.js	src/app/api/ai/settings/route.ts:43
POST	/api/github/validate	POST (route handler)	Not visible	api · Next.js	src/app/api/github/validate/route.ts:10
GET	/api/jobs/:id	GET (route handler)	Not visible	api · Next.js	src/app/api/jobs/[id]/route.ts:8
POST	/api/jobs/:id/cancel	POST (route handler)	Not visible	api · Next.js	src/app/api/jobs/[id]/cancel/route.ts:8
GET	/api/projects	GET (route handler)	Not visible	api · Next.js	src/app/api/projects/route.ts:7

Method	Route	Handler	Authentication	Kind	Location
DELETE	/api/projects/:id	DELETE (route handler)	Not visible	api · Next.js	src/app/api/projects/[id]/route.ts:15
GET	/api/projects/:id	GET (route handler)	Not visible	api · Next.js	src/app/api/projects/[id]/route.ts:8
POST	/api/projects/:id/analyze	POST (route handler)	Not visible	api · Next.js	src/app/api/projects/[id]/analyze/route.ts:17
GET	/api/projects/:id/architecture	GET (route handler)	Not visible	api · Next.js	src/app/api/projects/[id]/architecture/route.ts:8
GET	/api/projects/:id/ask	GET (route handler)	Not visible	api · Next.js	src/app/api/projects/[id]/ask/route.ts:11
POST	/api/projects/:id/ask	POST (route handler)	Not visible	api · Next.js	src/app/api/projects/[id]/ask/route.ts:19
GET	/api/projects/:id/deck	GET (route handler)	Not visible	api · Next.js	src/app/api/projects/[id]/deck/route.ts:14
GET	/api/projects/:id/explain	GET (route handler)	Not visible	api · Next.js	src/app/api/projects/[id]/explain/route.ts:27
POST	/api/projects/:id/explain	POST (route handler)	Not visible	api · Next.js	src/app/api/projects/[id]/explain/route.ts:48
GET	/api/projects/:id/export	GET (route handler)	Not visible	api · Next.js	src/app/api/projects/[id]/export/route.ts:16
GET	/api/projects/:id/files	GET (route handler)	Not visible	api · Next.js	src/app/api/projects/[id]/files/route.ts:9
GET	/api/projects/:id/files/content	GET (route handler)	Not visible	api · Next.js	src/app/api/projects/[id]/files/content/route.ts:15
GET	/api/projects/:id/findings	GET (route handler)	Not visible	api · Next.js	src/app/api/projects/[id]/findings/route.ts:13
GET	/api/projects/:id/formal	GET (route handler)	Not visible	api · Next.js	src/app/api/projects/[id]/formal/route.ts:8
GET	/api/projects/:id/graph	GET (route handler)	Not visible	api · Next.js	src/app/api/projects/[id]/graph/route.ts:9
GET	/api/projects/:id/impact	GET (route handler)	Not visible	api · Next.js	src/app/api/projects/[id]/impact/route.ts:8
GET	/api/projects/:id/overview	GET (route handler)	Not visible	api · Next.js	src/app/api/projects/[id]/overview/route.ts:12
GET	/api/projects/:id/report	GET (route handler)	Not visible	api · Next.js	src/app/api/projects/[id]/report/route.ts:12
GET	/api/projects/:id/search	GET (route handler)	Not visible	api · Next.js	src/app/api/projects/[id]/search/route.ts:8
GET	/api/projects/:id/symbols/:symbolId	GET (route handler)	Not visible	api · Next.js	src/app/api/projects/[id]/symbols/[symbolId]/route.ts:13
POST	/api/projects/github	POST (route handler)	Not visible	api · Next.js	src/app/api/projects/github/route.ts:16
POST	/api/projects/import-bundle	POST (route handler)	Not visible	api · Next.js	src/app/api/projects/import-bundle/route.ts:16
POST	/api/projects/upload	POST (route handler)	Not visible	api · Next.js	src/app/api/projects/upload/route.ts:16
GET	/api/status	GET (route handler)	Not visible	api · Next.js	src/app/api/status/route.ts:32

Method	Route	Handler	Authentication	Kind	Location
GET	/p/:id	OverviewPage	Not visible	page · Next.js	src/app/p/[id]/page.tsx:27
GET	/p/:id/architecture	ArchitecturePage	Not visible	page · Next.js	src/app/p/[id]/architecture/page.tsx:16
GET	/p/:id/ask	Cite	Not visible	page · Next.js	src/app/p/[id]/ask/page.tsx:20
GET	/p/:id/deck	DeckPage	Not visible	page · Next.js	src/app/p/[id]/deck/page.tsx:11
GET	/p/:id/explain	Origin	Not visible	page · Next.js	src/app/p/[id]/explain/page.tsx:13
GET	/p/:id/files	Splitter	Not visible	page · Next.js	src/app/p/[id]/files/page.tsx:36
GET	/p/:id/map	ImpactPanel	Not visible	page · Next.js	src/app/p/[id]/map/page.tsx:19
GET	/p/:id/proofs	Property	Not visible	page · Next.js	src/app/p/[id]/proofs/page.tsx:20

8. Data Architecture

12 data model(s) were found using Drizzle + SQL table projects (drizzle/0000_robust_hawkeye.sql:117), Drizzle + SQL table credentials (drizzle/0000_robust_hawkeye.sql:7), Drizzle + SQL table blobs (drizzle/0000_robust_hawkeye.sql:1), Drizzle + SQL table files (drizzle/0000_robust_hawkeye.sql:20) and 8 more. (src/lib/db/schema.ts:4-26, src/lib/db/schema.ts:29-33, src/lib/db/schema.ts:36-40)

9. External Integrations

Service	Category	Where used	Why	If unavailable
Anthropic	ai	package.json, src/lib/ai/anthropic.ts:1, src/lib/config.ts:51	language model inference	AI-generated features stop producing output
OpenAI	ai	src/lib/config.ts:61, src/lib/config.ts:60, src/lib/config.ts:49	language model inference and embeddings	AI-generated features stop producing output
SQLite	database	package.json, src/lib/db/client.ts:1	embedded relational storage	persisted reads and writes fail
GitHub API	developer-platform	src/lib/config.ts:81, src/lib/config.ts:80	repository and identity integration	GitHub-backed features fail

10. Infrastructure & Deployment

Deployment and operations assets: GitHub Actions, Docker, Docker Compose. (.github/workflows/ci.yml, Dockerfile, docker-compose.yml)

Network ports referenced: 3003, 3211, 4599. (Dockerfile:58, playwright.config.ts:5)

72 environment variable(s) configure the application, 6 of which look sensitive (names only: ANTHROPIC_API_KEY, BRODY_URL, CREDENTIAL_SECRET, GITHUB_TOKEN, OPENAI_API_KEY, OPENAI_BASE_URL).

Configuration (variable names only; values are never exported):

Variable	Sensitive	Read in	Declared in
AI_CONCURRENCY	no		.env.example
AI_EMBEDDING_MODEL	no	src/lib/config.ts:62	.env.example
AI_MAX_FILES_EXPLAINED	no		.env.example

Variable	Sensitive	Read in	Declared in
AI_MAX_FILES_REVIEWED	no		.env.example
AI_MAX_SYMBOLS_EXPLAINED	no		.env.example
AI_MODEL	no	src/lib/config.ts:50	
AI_PRICING	no	src/lib/ai/pricing.ts:62	
AI_PROVIDER	no	src/lib/config.ts:42	.env.example
AI_REFUSAL_FALLBACKS	no	src/lib/ai/anthropic.ts:33	.env.example
AI_SETTINGS_PATH	no	src/lib/ai/settings.ts:40	
ALLOWED_HOSTS	no	src/proxy.ts:9	.env.example, docker-compose.yml
ANTHROPIC_API_KEY	yes	src/lib/config.ts:51	.env.example
ANTHROPIC_MODEL	no	src/lib/config.ts:48	
ANTHROPIC_WORKSPACE_ID	no	src/lib/config.ts:53	.env.example
BRODY_BROWSER	no	scripts/brody.sh:47	
BRODY_HOST	no	scripts/brody.sh:18	
BRODY_OPEN	no	scripts/brody.sh:42	
BRODY_PARSE_WORKER	no	src/lib/config.ts:93	
BRODY_URL	yes	src/lib/bundle/launchers.ts:45	
BUNDLE_VERSION	no	src/lib/bundle/index.ts:209	
CLOSE_TAG	no	src/lib/ai/prompt.ts:30	
CREDENTIAL_SECRET	yes	src/lib/config.ts:124	.env.example
DATABASE_PATH	no	drizzle.config.ts:7, src/lib/config.ts:24	.env.example, docker-compose.yml
E2E MOCK_AI_PORT	no	playwright.config.ts:26	
E2E_PORT	no	playwright.config.ts:5	
ELAN_HOME	no	src/lib/formal/lean.ts:97	Dockerfile
EMBEDDED_WORKER	no	src/instrumentation.ts:3	.env.example
FORMAL_CONCURRENCY	no		.env.example
FORMAL_MAX_HEARTBEATS	no		.env.example
FORMAL_MAX_REPAIR_ROUNDS	no		.env.example
FORMAL_MAX_TARGETS	no		.env.example
FORMAL_MEMORY_MB	no		.env.example
FORMAL_SANDBOX	no	src/lib/config.ts:121	
FORMAL_TIMEOUT_MS	no		.env.example
FORMAL_VERIFICATION	no	src/lib/config.ts:108	.env.example
FORMAL_VERSION	no	src/lib/formal/index.ts:259	
GITHUB_API_BASE	no	src/lib/config.ts:81	
GITHUB_TOKEN	yes	src/lib/config.ts:80	.env.example
GO_PATH	no	src/lib/config.ts:99	
LEAN_BIN	no	src/lib/config.ts:109	Dockerfile

Variable	Sensitive	Read in	Declared in
LEAN_RULES	no	src/lib/formal/index.ts:110	
LEAN_TOOLCHAIN	no		Dockerfile
MAX_FILE_BYTES	no		.env.example
MAX_FILES	no		.env.example
MAX_TOTAL_BYTES	no		.env.example
MAX_UPLOAD_BYTES	no	next.config.ts:17	.env.example
MAX_ZIP_ENTRIES	no		.env.example
MAX_ZIP_RATIO	no		.env.example
MENU_WIDTH	no	src/components/download.tsx:104	
MERMAID_ONLOAD	no	src/lib/export/html.ts:86	
NEXT_BUILD_CPUS	no	next.config.ts:19	
NEXT_RUNTIME	no	src/instrumentation.ts:3	
NEXT_TELEMETRY_DISABLED	no		Dockerfile
NODE_ENV	no		Dockerfile
OPEN_TAG	no	src/lib/ai/prompt.ts:30	
OPENAI_API_KEY	yes	src/lib/config.ts:61	.env.example
OPENAI_BASE_URL	yes	src/lib/config.ts:60	.env.example
OPENAI_MODEL	no	src/lib/config.ts:49	
PARSE_WORKER_MIN_FILES	no	src/lib/config.ts:92	
PARSE_WORKERS	no	src/lib/config.ts:91	

11. Testing Strategy

60 test file(s): 13 e2e, 17 fixture, 27 unit, 1 integration, 2 helper; frameworks: Vitest, Playwright. (e2e/ai-settings.spec.ts, e2e/bundle.spec.ts)

Important files with no test referencing them include src/components/ui.tsx. (src/components/ui.tsx)

12. Security Model

Authentication: no mechanism detected. 33 of 33 API route(s) have no visible access check.

Secrets: 11 likely secret value(s) were found in source (redacted in this report). (e2e/bundle.spec.ts:77, fixtures/sample-shop/src/config.ts:7)

8 security finding(s) were recorded: 0 critical/high.

13. Code Review

25 finding(s): 0 critical, 0 high, 8 medium, 17 low, 0 informational. 25 from static analyzers, 0 AI-inferred (0 need verification).

By category: Maintainability 16 · Security 8 · Testing 1

ID	Severity	Finding	Where	Fix
MNT-001	Medium	High cyclomatic complexity in Home (110)	src/app/page.tsx:52	Simplify with early returns, lookup tables or smaller functions, and add tests covering each branch.

ID	Severity	Finding	Where	Fix
MNT-002	Medium	High cyclomatic complexity in buildDeckContent (123)	src/lib/deck/content.ts:88	Simplify with early returns, lookup tables or smaller functions, and add tests covering each branch.
MNT-003	Medium	High cyclomatic complexity in detectRoutes (190)	src/lib/discover/detect.ts:250	Simplify with early returns, lookup tables or smaller functions, and add tests covering each branch.
MNT-004	Medium	High cyclomatic complexity in detectEntryPoints (100)	src/lib/discover/detect.ts:492	Simplify with early returns, lookup tables or smaller functions, and add tests covering each branch.
MNT-005	Medium	High cyclomatic complexity in enhanceWithAI (107)	src/lib/docs/ai.ts:80	Simplify with early returns, lookup tables or smaller functions, and add tests covering each branch.
MNT-006	Medium	High cyclomatic complexity in buildMarkdown (109)	src/lib/export/markdown.ts:63	Simplify with early returns, lookup tables or smaller functions, and add tests covering each branch.
MNT-007	Medium	High cyclomatic complexity in buildGraph (156)	src/lib/graph/build.ts:70	Simplify with early returns, lookup tables or smaller functions, and add tests covering each branch.
MNT-008	Medium	High cyclomatic complexity in extractTsJs (166)	src/lib/parse/extract.ts:166	Simplify with early returns, lookup tables or smaller functions, and add tests covering each branch.
MNT-009	Low	Very long component: Home (278 lines)	src/app/page.tsx:52	Extract cohesive steps into named helper functions.
MNT-010	Low	Very long function: buildDeckContent (248 lines)	src/lib/deck/content.ts:88	Extract cohesive steps into named helper functions.
MNT-011	Low	Very long function: detectRoutes (190 lines)	src/lib/discover/detect.ts:250	Extract cohesive steps into named helper functions.
MNT-012	Low	Very long function: enhanceWithAI (244 lines)	src/lib/docs/ai.ts:80	Extract cohesive steps into named helper functions.
MNT-013	Low	Very long function: buildMarkdown (199 lines)	src/lib/export/markdown.ts:63	Extract cohesive steps into named helper functions.
MNT-014	Low	Very long function: buildGraph (292 lines)	src/lib/graph/build.ts:70	Extract cohesive steps into named helper functions.
MNT-015	Low	Very long function: layoutMap (185 lines)	src/lib/map/layout.ts:71	Extract cohesive steps into named helper functions.
MNT-016	Low	Very long function: extractTsJs (198 lines)	src/lib/parse/extract.ts:166	Extract cohesive steps into named helper functions.
SEC-001	Low	Secret-like value in test or fixture file e2e/bundle.spec.ts	e2e/bundle.spec.ts:77	Rotate the credential immediately, remove it from history, and load it from environment configuration or a...
SEC-002	Low	Secret-like value in test or fixture file fixtures/sample-shop/src/config.ts	fixtures/sample-shop/src/config.ts:7	Rotate the credential immediately, remove it from history, and load it from environment configuration or a...
SEC-003	Low	None of the 33 API routes has an authentication check	src/app/api/ai/models/route.ts:9	Put the service behind an authenticating reverse proxy, or add authentication and authorization to the...
SEC-004	Low	Secret-like value in test or fixture file tests/ai.test.ts	tests/ai.test.ts:158	Rotate the credential immediately, remove it from history, and load it from environment configuration or a...
SEC-005	Low	Secret-like value in test or fixture file tests/api.test.ts	tests/api.test.ts:117	Rotate the credential immediately, remove it from history, and load it from environment configuration or a...

ID	Severity	Finding	Where	Fix
SEC-006	Low	Secret-like value in test or fixture file tests/ bundle.test.ts	tests/ bundle.test.ts:26	Rotate the credential immediately, remove it from history, and load it from environment configuration or a...
SEC-007	Low	Secret-like value in test or fixture file tests/ pipeline.test.ts	tests/ pipeline.test.ts:87	Rotate the credential immediately, remove it from history, and load it from environment configuration or a...
SEC-008	Low	Secret-like value in test or fixture file tests/ secrets.test.ts	tests/secrets.test.ts:5	Rotate the credential immediately, remove it from history, and load it from environment configuration or a...
TST-001	Low	1 important file has no tests	src/components/ ui.tsx:1	Add unit tests for the exported functions and failure paths of these files, starting with the request...

14. Engineering Risks

No critical or high-severity findings were recorded.

15. File & Component Explanations

Each entry explains one file on its own. How groups of files work together is in *Major Functional Areas and Folder & Module Explanations*.

File	Role	What it does
src/lib/db/client.ts	data	data-layer TypeScript file in the Data Layer area. It declares 14 functions, 5 variables, 3 types, 1 interface, 1 constant.
src/lib/config.ts	service	service-layer TypeScript file in the Shared Utilities area. It declares 1 variable, 1 type, 2 functions.
src/lib/db/schema.ts	schema	TypeScript file in the Data Layer area. It defines the data models projects, credentials, blobs, files and 8 more. It declares 12 models, 7 types, 1...
src/lib/api.ts	api	API layer TypeScript file in the Shared Utilities area. It declares 7 functions.
src/lib/util/errors.ts	util	TypeScript file in the Util area. It declares 1 class, 2 functions, 3 propertyts, 1 method.
src/lib/ai/index.ts	ai	TypeScript file in the AI Processing area. It declares 7 functions, 2 interfaces, 1 class, 1 method, 3 variables, 2 propertyts.
src/lib/discover/types.ts	util	TypeScript file in the Discover area. It declares 17 interfaces.
src/lib/docs/types.ts	util	TypeScript file in the Documentation area. It declares 11 interfaces.
src/lib/graph/build.ts	service	service-layer TypeScript file in the Graph area. It declares 4 interfaces, 3 functions, 3 constants.
.github/workflows/ci.yml	infra	infrastructure YAML file in the Infrastructure & Deployment area. It declares 3 jobs.
src/lib/ai/models.ts	ai	TypeScript file in the AI Processing area. It declares 6 functions, 2 constants, 1 type, 1 interface, 1 variable.
src/lib/docs/ai.ts	service	service-layer TypeScript file in the Documentation area. It declares 7 functions, 1 interface, 9 variables, 1 constant.
src/lib/ingest/store.ts	service	service-layer TypeScript file in the Content Ingestion area. It declares 8 functions, 1 interface, 1 type, 1 constant, 2 variables.
src/lib/map/index.ts	service	service-layer TypeScript file in the Map area. It declares 24 functions, 8 interfaces, 2 variables, 2 constants.
src/lib/jobs/index.ts	job	background-processing TypeScript file in the Background Jobs area. It declares 9 functions, 13 methods, 2 classs, 5 propertyts, 1 constant.

The 15 most important of 191 documented files. The *Complete Technical Report* explains every file.

16. Detailed Symbol Explanations

Symbol	Kind	Purpose	Used by
jobs.push	job	Job jobs.push (no doc comment; role inferred from name and relationships).	10 callers
split	function	Function split (no doc comment; role inferred from name and relationships).	10 callers
trim	function	Function trim (no doc comment; role inferred from name and relationships).	10 callers
getDb	function	Return the open database, opening the configured one on first use.	10 callers
openDatabase	function	Function openDatabase (no doc comment; role inferred from name and relationships).	5 callers
migrationsFolder	function	Resolve the migrations folder for both source and standalone builds.	1 caller
files	model	Data model files.	10 callers
json	function	Function json (no doc comment; role inferred from name and relationships).	10 callers
projects	model	Projects: one row per ingested codebase (a repository or an upload).	10 callers
AppError	class	Application error with an HTTP status and a user-facing, actionable message.	10 callers
jobs	model	Data model jobs.	10 callers
findings	model	Data model findings.	10 callers
symbols	model	Data model symbols.	10 callers
splitColor	function	"#RRGGBB" or "#RRGGBBAA" split into the opaque colour and its alpha, for formats that keep them apart.	8 callers
jobs.verify	job	Job jobs.verify (no doc comment; role inferred from name and relationships).	1 caller

The 15 most important of 300 documented symbols. The Complete Technical Report explains every one.

17. Recommendations

1. Add tests for src/components/ui.tsx, which many files depend on. (src/components/ui.tsx)

18. Detailed Code Map

1. Repository Tree

```
brody/
├── .github/ # build, deployment and operations (1 files)
│   └── workflows/ # infrastructure (1 files)
├── docs/ # project documentation (7 files)
│   ├── validation/
│   └── ARCHITECTURE.md
├── drizzle/
│   ├── meta/ # configuration (2 files)
│   └── 0000_robust_hawkeye.sql
├── e2e/ ○ # automated tests, fixtures and mocks (13 files)
│   ├── ai-settings.spec.ts
│   ├── bundle.spec.ts ○
│   ├── datamodel.spec.ts
│   ├── deck.spec.ts
│   ├── explain.spec.ts
│   ├── home.spec.ts
│   ├── performance.spec.ts
│   ├── progress.spec.ts
│   ├── proofs.spec.ts
│   ├── styling.spec.ts
│   └── ... 3 more
└── fixtures/ ○ # automated tests, fixtures and mocks (17 files)
```

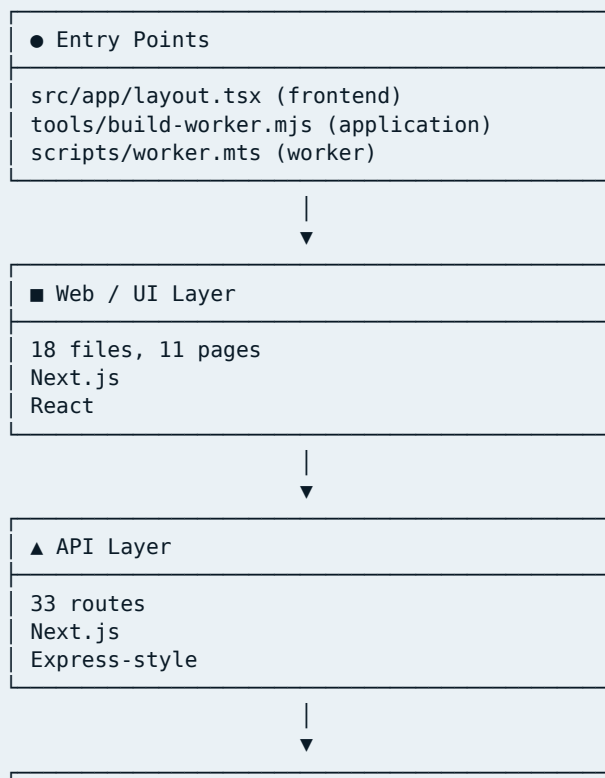


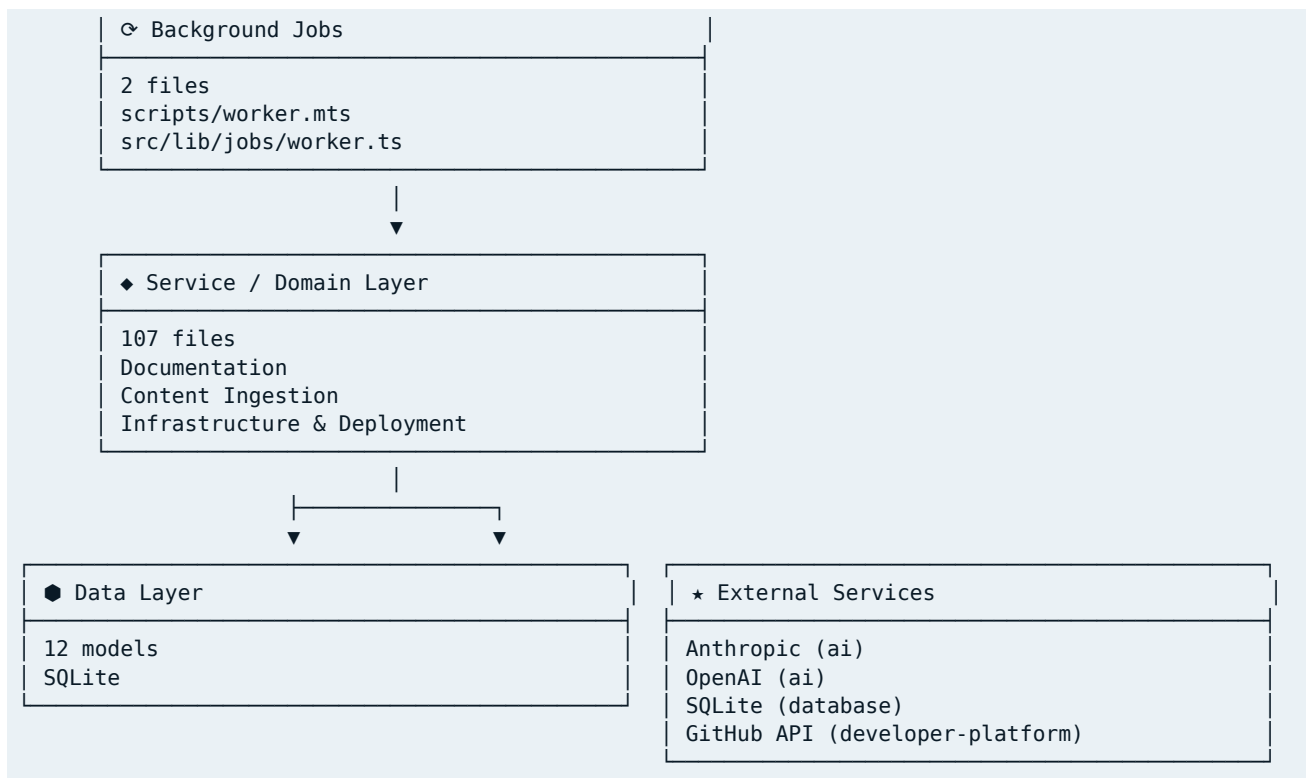
```

├── sample-shop/ ○ # tests (17 files)
├── public/ # unknown (5 files)
│   ├── file.svg
│   ├── globe.svg
│   ├── next.svg
│   ├── vercel.svg
│   └── window.svg
├── scripts/ # build, deployment and operations (7 files)
│   ├── analyze.mts
│   ├── benchmark.mts
│   ├── brody.sh
│   ├── convert.mts
│   ├── validate.sh
│   ├── verify-ai.mts
│   └── worker.mts
├── src/ ●
│   ├── app/ ● # API layer (42 files)
│   ├── components/ ○ # pages, components and client-side state (14 files)
│   ├── lib/ ● # reusable helpers and shared code (105 files)
│   ├── instrumentation.ts
│   └── proxy.ts
├── tests/ ○ # automated tests, fixtures and mocks (30 files)
│   ├── helpers/ # reusable helpers and shared code (2 files)
│   ├── ai.test.ts ○
│   ├── analyzers.test.ts
│   ├── api.test.ts ○
│   ├── brief.test.ts
│   ├── bundle.test.ts ○
│   ├── datamodel.test.ts
│   ├── deck.test.ts
│   ├── dual-provider.test.ts
│   ├── exports.test.ts
│   └── ... 19 more
├── tools/ # build, deployment and operations (1 files)
│   └── build-worker.mjs
└── ... 17 more

```

2. Functional Component Map





Functional-area dependencies:

- API Layer → Shared Utilities (186 references)
- API Layer → User Interface (176 references)
- Testing → Data Layer (120 references)
- Testing → AI Processing (115 references)
- Testing → Content Ingestion (68 references)
- Testing → Shared Utilities (55 references)
- API Layer → Documentation (51 references)
- API Layer → Map (49 references)
- Testing → Background Jobs (47 references)
- API Layer → Util (44 references)

3. Runtime / Data Flow Map

```

GET /path
  ○ GET /path (src/lib/discover/detect.ts:250)
    ↓
    ○ lineAt (src/lib/discover/detect.ts:6)
      ↓
      ○ callArgs (src/lib/discover/detect.ts:218)
        ↓
        ○ isInlineFn (src/lib/discover/detect.ts:248)
          ↓
          ● reads symbols (src/lib/db/schema.ts:82)
            ↓
            ○ jobs.push (.github/workflows/ci.yml:3)
              ↓
              ○ split (src/lib/ai/models.ts:67)
                ↓
                ○ trim (src/lib/docs/ai.ts:404)
  
```

```

GET /api/ai/models
  ○ GET /api/ai/models (src/app/api/ai/models/route.ts:9)
    ↓
  
```

- guard (src/lib/api.ts:20)
- ↓
- AppError (src/lib/util/errors.ts:2)
- ↓
- json (src/lib/api.ts:8)
- ↓
- listModels (src/lib/ai/models.ts:76)
- ↓
- fail (src/lib/api.ts:12)
- ↓
- resolveModel (src/lib/ai/settings.ts:135)
- ↓
- providerConfigured (src/lib/ai/settings.ts:105)
- ↓
- explainAIError (src/lib/ai/errors.ts:2)
- ↓
- cacheKey (src/lib/ai/models.ts:53)
- ↓
- split (src/lib/ai/models.ts:67)
- ↓
- fetchCatalogue (src/lib/ai/models.ts:62)
- ↓
- isAppError (src/lib/util/errors.ts:14)
- ↓
- scrubToken (src/lib/ingest/credentials.ts:36)

GET /api/ai/settings

- GET /api/ai/settings (src/app/api/ai/settings/route.ts:38)
- ↓
- guard (src/lib/api.ts:20)
- ↓
- json (src/lib/api.ts:8)
- ↓
- view (src/app/api/ai/settings/route.ts:21)
- ↓
- fail (src/lib/api.ts:12)
- ↓
- readSettings (src/lib/ai/settings.ts:48)
- ↓
- providerStatus (src/lib/ai/index.ts:56)
- ↓
- providerConfigured (src/lib/ai/settings.ts:105)
- ↓
- resolveModel (src/lib/ai/settings.ts:135)
- ↓
- runtimeKind (src/lib/runtime.ts:5)
- ↓
- resolveEmbeddingModel (src/lib/ai/settings.ts:149)
- ↓
- endpoint (src/app/api/ai/settings/route.ts:11)
- ↓
- selectedProvider (src/lib/ai/settings.ts:120)
- ↓
- isAppError (src/lib/util/errors.ts:14)

PUT /api/ai/settings

- PUT /api/ai/settings (src/app/api/ai/settings/route.ts:43)
- ↓
- guard (src/lib/api.ts:20)
- ↓
- AppError (src/lib/util/errors.ts:2)
- ↓
- json (src/lib/api.ts:8)
- ↓
- updateSettings (src/lib/ai/settings.ts:63)
- ↓
- resetProviderCache (src/lib/ai/index.ts:35)

```

↓
○ checkProvider (src/lib/ai/index.ts:108)
↓
○ view (src/app/api/ai/settings/route.ts:21)
↓
○ fail (src/lib/api.ts:12)
↓
○ readSettings (src/lib/ai/settings.ts:48)
↓
○ settingsPath (src/lib/ai/settings.ts:39)
↓
○ withDeadline (src/lib/ai/deadline.ts:2)
↓
○ explainAIError (src/lib/ai/errors.ts:2)
↓
○ providerStatus (src/lib/ai/index.ts:56)

```

4. Dependency Map

The 12 most important of 182 source files (the interactive map in the application shows all of them):

- client.ts
- config.ts
- schema.ts
- api.ts
- errors.ts
- index.ts
- types.ts
- types.ts
- build.ts
- models.ts
- ai.ts
- store.ts

5. API Map

Method	Route	Handler	Authentication	Kind	Location
GET	/	uploadWithProgress	Not visible	page · Next.js	src/app/page.tsx:20
GET	/api/ai/models	GET (route handler)	Not visible	api · Next.js	src/app/api/ai/models/route.ts:9
GET	/api/ai/settings	GET (route handler)	Not visible	api · Next.js	src/app/api/ai/settings/route.ts:38
PUT	/api/ai/settings	PUT (route handler)	Not visible	api · Next.js	src/app/api/ai/settings/route.ts:43
POST	/api/github/validate	POST (route handler)	Not visible	api · Next.js	src/app/api/github/validate/route.ts:10
GET	/api/jobs/:id	GET (route handler)	Not visible	api · Next.js	src/app/api/jobs/[id]/route.ts:8
POST	/api/jobs/:id/cancel	POST (route handler)	Not visible	api · Next.js	src/app/api/jobs/[id]/cancel/route.ts:8
GET	/api/projects	GET (route handler)	Not visible	api · Next.js	src/app/api/projects/route.ts:7
DELETE	/api/projects/:id	DELETE (route handler)	Not visible	api · Next.js	src/app/api/projects/[id]/route.ts:15

Method	Route	Handler	Authentication	Kind	Location
GET	/api/projects/:id	GET (route handler)	Not visible	api · Next.js	src/app/api/projects/[id]/route.ts:8
POST	/api/projects/:id/analyze	POST (route handler)	Not visible	api · Next.js	src/app/api/projects/[id]/analyze/route.ts:17
GET	/api/projects/:id/architecture	GET (route handler)	Not visible	api · Next.js	src/app/api/projects/[id]/architecture/route.ts:8
GET	/api/projects/:id/ask	GET (route handler)	Not visible	api · Next.js	src/app/api/projects/[id]/ask/route.ts:11
POST	/api/projects/:id/ask	POST (route handler)	Not visible	api · Next.js	src/app/api/projects/[id]/ask/route.ts:19
GET	/api/projects/:id/deck	GET (route handler)	Not visible	api · Next.js	src/app/api/projects/[id]/deck/route.ts:14
GET	/api/projects/:id/explain	GET (route handler)	Not visible	api · Next.js	src/app/api/projects/[id]/explain/route.ts:27
POST	/api/projects/:id/explain	POST (route handler)	Not visible	api · Next.js	src/app/api/projects/[id]/explain/route.ts:48
GET	/api/projects/:id/export	GET (route handler)	Not visible	api · Next.js	src/app/api/projects/[id]/export/route.ts:16
GET	/api/projects/:id/files	GET (route handler)	Not visible	api · Next.js	src/app/api/projects/[id]/files/route.ts:9
GET	/api/projects/:id/files/content	GET (route handler)	Not visible	api · Next.js	src/app/api/projects/[id]/files/content/route.ts:15
GET	/api/projects/:id/findings	GET (route handler)	Not visible	api · Next.js	src/app/api/projects/[id]/findings/route.ts:13
GET	/api/projects/:id/formal	GET (route handler)	Not visible	api · Next.js	src/app/api/projects/[id]/formal/route.ts:8
GET	/api/projects/:id/graph	GET (route handler)	Not visible	api · Next.js	src/app/api/projects/[id]/graph/route.ts:9
GET	/api/projects/:id/impact	GET (route handler)	Not visible	api · Next.js	src/app/api/projects/[id]/impact/route.ts:8
GET	/api/projects/:id/overview	GET (route handler)	Not visible	api · Next.js	src/app/api/projects/[id]/overview/route.ts:12
GET	/api/projects/:id/report	GET (route handler)	Not visible	api · Next.js	src/app/api/projects/[id]/report/route.ts:12
GET	/api/projects/:id/search	GET (route handler)	Not visible	api · Next.js	src/app/api/projects/[id]/search/route.ts:8
GET	/api/projects/:id/symbols/:symbolId	GET (route handler)	Not visible	api · Next.js	src/app/api/projects/[id]/symbols/[symbolId]/route.ts:13
POST	/api/projects/github	POST (route handler)	Not visible	api · Next.js	src/app/api/projects/github/route.ts:16
POST	/api/projects/import-bundle	POST (route handler)	Not visible	api · Next.js	src/app/api/projects/import-bundle/route.ts:16
POST	/api/projects/upload	POST (route handler)	Not visible	api · Next.js	src/app/api/projects/upload/route.ts:16
GET	/api/status	GET (route handler)	Not visible	api · Next.js	src/app/api/status/route.ts:32
GET	/p/:id	OverviewPage	Not visible	page · Next.js	src/app/p/[id]/page.tsx:27

Method	Route	Handler	Authentication	Kind	Location
GET	/p/:id/architecture	ArchitecturePage	Not visible	page · Next.js	src/app/p/[id]/architecture/page.tsx:16
GET	/p/:id/ask	Cite	Not visible	page · Next.js	src/app/p/[id]/ask/page.tsx:20
GET	/p/:id/deck	DeckPage	Not visible	page · Next.js	src/app/p/[id]/deck/page.tsx:11
GET	/p/:id/explain	Origin	Not visible	page · Next.js	src/app/p/[id]/explain/page.tsx:13
GET	/p/:id/files	Splitter	Not visible	page · Next.js	src/app/p/[id]/files/page.tsx:36
GET	/p/:id/map	ImpactPanel	Not visible	page · Next.js	src/app/p/[id]/map/page.tsx:19
GET	/p/:id/proofs	Property	Not visible	page · Next.js	src/app/p/[id]/proofs/page.tsx:20
GET	/p/:id/reports	ReportsPage	Not visible	page · Next.js	src/app/p/[id]/reports/page.tsx:18
GET	/p/:id/review	Diff	Not visible	page · Next.js	src/app/p/[id]/review/page.tsx:14
GET	/path	handler	Not visible	api · Express-style	src/lib/discover/detect.ts:315
GET	/x	GET /x (inline)	Not visible	api · Express-style	src/lib/parse/extract.ts:328

6. Data Model Map

Model	Type	Fields	Points to	Location
projects	Drizzle + SQL table projects (drizzle/0000_robust_hawkeye.sql:117)	id: text("id").primaryKey(), name: text("name").notNull(), sourceType: text("source_type").notNull(), sourceUrl: text("source_url"), owner: text("owner"), branch: text("branch")	—	src/lib/db/schema.ts:4
credentials	Drizzle + SQL table credentials (drizzle/0000_robust_hawkeye.sql:7)	projectId: text("project_id").primaryKey(), encrypted: text("encrypted").notNull(), createdAt: integer("created_at").notNull(), project_id, created_at	—	src/lib/db/schema.ts:29
blobs	Drizzle + SQL table blobs (drizzle/0000_robust_hawkeye.sql:1)	hash: text("hash").primaryKey(), content: text("content").notNull(), size: integer("size").notNull()	—	src/lib/db/schema.ts:36
files	Drizzle + SQL table files (drizzle/0000_robust_hawkeye.sql:20)	id: text("id").primaryKey(), projectId: text("project_id").notNull(), path: text("path").notNull(), name: text("name").notNull(), directory: text("directory").notNull(), extension: text("extension").notNull()	—	src/lib/db/schema.ts:42
symbols	Drizzle + SQL table symbols (drizzle/0000_robust_hawkeye.sql:165)	id: text("id").primaryKey(), projectId: text("project_id").notNull(), fileId: text("file_id").notNull(), filePath: text("file_path").notNull(), name: text("name").notNull(), qualifiedName: text("qualified_name").notNull()	—	src/lib/db/schema.ts:82

Model	Type	Fields	Points to	Location
relationships	Drizzle + SQL table relationships (drizzle/0000_robust_hawkeye.sql:148)	id: text("id").primaryKey(), projectId: text("project_id").notNull(), kind: text("kind").notNull(), sourceType: text("source_type").notNull(), sourceId: text("source_id").notNull(), targetType: text("target_type").notNull()	—	src/lib/db/schema.ts:110
findings	Drizzle + SQL table findings (drizzle/0000_robust_hawkeye.sql:54)	id: text("id").primaryKey(), projectId: text("project_id").notNull(), code: text("code").notNull(), title: text("title").notNull(), category: text("category").notNull(), severity: text("severity").notNull()	—	src/lib/db/schema.ts:128
jobs	Drizzle + SQL table jobs (drizzle/0000_robust_hawkeye.sql:93)	id: text("id").primaryKey(), projectId: text("project_id").notNull(), status: text("status").notNull(), currentStage: text("current_stage"), error: text("error"), createdAt: integer("created_at").notNull()	—	src/lib/db/schema.ts:159
parseCache	Drizzle + SQL table parse_cache (drizzle/0000_robust_hawkeye.sql:111)	key: text("key").primaryKey(), payload: text("payload").notNull(), createdAt: integer("created_at").notNull(), created_at	—	src/lib/db/schema.ts:182
explainCache	Drizzle + SQL table explain_cache (drizzle/0000_robust_hawkeye.sql:13)	key: text("key").primaryKey(), kind: text("kind").notNull(), createdAt: integer("created_at").notNull(), payload, created_at	—	src/lib/db/schema.ts:189
indexEntries	Drizzle + SQL table index_entries (drizzle/0000_robust_hawkeye.sql:80)	projectId: text("project_id").notNull(), kind: text("kind").notNull(), refId: text("ref_id").notNull(), filePath: text("file_path"), title: text("title").notNull(), terms: text("terms").notNull()	—	src/lib/db/schema.ts:197
questions	Drizzle + SQL table questions (drizzle/0000_robust_hawkeye.sql:139)	id: text("id").primaryKey(), projectId: text("project_id").notNull(), question: text("question").notNull(), createdAt: integer("created_at").notNull(), answer, project_id	—	src/lib/db/schema.ts:212

7. Test Map

Test file	Kind	Framework	Exercises
e2e/ai-settings.spec.ts	e2e	Playwright	.github/workflows/ci.yml, src/lib/deck/pdf.ts, src/lib/api.ts, src/components/ai-settings.tsx
e2e/bundle.spec.ts	e2e	Playwright	src/lib/ai/models.ts, src/lib/api.ts
e2e/datamodel.spec.ts	e2e	Playwright	not resolved
e2e/deck.spec.ts	e2e	Playwright	src/lib/ai/models.ts, src/lib/api.ts, src/components/deck.tsx
e2e/explain.spec.ts	e2e	Playwright	src/lib/deck/pdf.ts
e2e/home.spec.ts	e2e	Playwright	src/lib/deck/pdf.ts
e2e/performance.spec.ts	e2e	Playwright	.github/workflows/ci.yml, src/lib/jobs/index.ts
e2e/progress.spec.ts	e2e	Playwright	src/lib/ai/openai.ts, src/lib/ai/models.ts, src/lib/api.ts
e2e/proofs.spec.ts	e2e	Playwright	src/lib/ai/openai.ts, .github/workflows/ci.yml
e2e/styling.spec.ts	e2e	Playwright	.github/workflows/ci.yml
e2e/theme.spec.ts	e2e	Playwright	src/lib/deck/theme.ts
e2e/usage.spec.ts	e2e	Playwright	src/components/usage.tsx

Test file	Kind	Framework	Exercises
e2e/workflow.spec.ts	e2e	Playwright	src/lib/deck/pdf.ts
fixtures/sample-shop/.env.example	fixture		.env.example
fixtures/sample-shop/.gitignore	fixture		.dockerignore

Important components with no tests:

- `src/components/ui.tsx` — high-importance ui file with no test referencing it

8. External Integration Map

```

★ Anthropic (ai)
  ↳ package.json, src/lib/ai/anthropic.ts:1, src/lib/config.ts:51
★ OpenAI (ai)
  ↳ src/lib/config.ts:61, src/lib/config.ts:60, src/lib/config.ts:49
★ SQLite (database)
  ↳ package.json, src/lib/db/client.ts:1
★ GitHub API (developer-platform)
  ↳ src/lib/config.ts:81, src/lib/config.ts:80

```

9. High-Risk Component Map

Risk	File	Role	Importance	Findings
● medium	src/lib/graph/build.ts	service	0.31	2
● medium	src/lib/docs/ai.ts	service	0.29	2
● medium	src/lib/export/markdown.ts	service	0.08	2
● medium	src/lib/discover/detect.ts	api	0.06	3
● medium	src/lib/deck/content.ts	service	0.05	2
● medium	src/lib/parse/extract.ts	api	0.05	2
● medium	src/app/page.tsx	api	0.03	2
○ low	src/components/ui.tsx	ui	0.19	1

10. Change Impact Relationships

split (`src/lib/ai/models.ts:67`) — Changing split can affect 201 upstream components, including 8 routes, and depends on 0 downstream components. 25 test file(s) exercise this code.

```

split
  ↓ used by
GET

split
  ↓ used by
GET

split
  ↓ used by
Cite

split
  ↓ used by
CollectionBody

split
  ↓ used by
Code

```


trim (src/lib/docs/ai.ts:404) — Changing trim can affect 201 upstream components, including 12 routes, and depends on 0 downstream components. 18 test file(s) exercise this code.

```
trim
  ↓ used by
POST

trim
  ↓ used by
GET

trim
  ↓ used by
BodySchema

trim
  ↓ used by
GET

trim
  ↓ used by
GET
```

11. Legend

LEGEND

- Application Entry Point
- ◆ Service
- UI Component
- ▲ API Endpoint
- Data Model
- Utility
- ★ External Service
- T Test
- C Configuration
- ⌚ Background Job
- ▣ Infrastructure
- Calls
- Imports
- Data flow
- ↳ Writes to
- ↵ Reads from
- ... Optional dependency
- Risk:
 - ! Critical
 - ▲ High
 - Medium
 - Low

19. Legend

LEGEND

- Application Entry Point
- ◆ Service
- UI Component
- ▲ API Endpoint
- Data Model
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- ★ External Service
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🔄 Background Job

■ Infrastructure

→ Calls

→ Imports

→ Data flow

↳ Writes to

↳ Reads from

... Optional dependency

Risk:

! Critical

▲ High

● Medium

○ Low